

Implementation of LSTM Optimized by Simulated Annealing for Toxicity Prediction: Case Study AR-LBD Toxicity

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Abstract—Over time, many individuals have been exposed to chemical substances with potentially harmful effects on the human body. Drug toxicity is a critical factor in drug development process. However, high toxicity remains a primary cause of drug failure during clinical trials. Therefore, toxicity testing has become a main focus in the medical field to prevent further exposure to hazardous chemicals. The study focuses on predicting the toxicity of androgen receptor ligand-binding domain (AR-LBD) compounds by enhancing the predictive capabilities of a Long Short-Term Memory (LSTM) model optimized by Simulated Annealing (SA). The methodology includes several steps, such as dataset preparation from the Tox21 Data Challenge, model training using the SA-optimized LSTM, performance evaluation against traditional toxicity prediction methods, and validation through testing datasets to confirm the reliability and effectiveness of the approach. The LSTM model, optimized by SA, demonstrated impressive results. On the test set, the proposed model outperforms traditional toxicity prediction models by achieving an F1-score of 0.7105 and an accuracy 0.9782. These results support the effectiveness of SA for optimization as it also outperformed baseline models where SA was not used. These findings indicate the importance of robust optimization techniques in improving predictive capabilities and applicabilities in real-world scenarios. Future research should explore additional hyperparameter tuning and alternative optimization techniques to further improve prediction accuracy and applicability across diverse datasets and various scenarios in toxicology.

Index Terms—toxicity prediction, LSTM, SA, AR-LBD

I. INTRODUCTION

In the medical field, a drug refers to a chemical substance that can be used to prevent, diagnose, treat, ease disease symptoms, and restore physiological functions in living organisms [1]–[3]. Over time, many individuals have been affected by chemical substances that pose potential harm to the human body. Drug toxicity is a critical aspect of drug development, along with the identification of adverse effects of substances on humans, plants, animals, and the environment. However, high toxicity remains a major factor contributing to drug failure in clinical trials [4]. Therefore, toxicity testing is an essential consideration in the medical field to minimize exposure to harmful chemical substances.

The increasing rates of drug withdrawals, rising costs, late-stage failures, and significant recalls represent some of the major challenges in drug toxicity testing. These issues are critical in pharmaceutical research and development (R&D). Data from CMR International demonstrated that between 2006 and 2010, approximately 22% of clinical development

failures and 54% of preclinical development failures were attributed to toxicity [5]. Therefore, in addition to minimizing exposure to harmful chemical substances, toxicity testing is a critical step in the early phases of drug research and development. This approach aims to prevent toxic drugs from progressing to clinical trials while also reducing the costs and labor associated with preclinical drug testing [6].

Traditionally, toxicity testing or measurement has been conducted through in vivo methods, where tests are performed directly on living organisms to observe their physiological reactions to the tested substances [7]. This approach is commonly used for toxicity assessments, with animals serving as the primary test subjects. However, this method is highly expensive and challenging to implement, requiring two to three years per chemical substance for comprehensive toxicity evaluation [8]. Additionally, traditional toxicity prediction methods often struggle with accurately classifying compounds, particularly the compounds with similar structures, leading to high rates of false positives and negatives. Moreover, the results from conventional in vivo testing offer limited insights into toxicity reactions in the human body due to interspecies differences and the diverse nature of disease models [6].

In response to this, the utilization of computational or in silico toxicity testing is recommended as it can reduce costs and uncertainties associated with animal testing [6], [8]. The in silico method offers several advantages, including cost reduction, greater time efficiency, and a unique approach to visualize toxicity testing results through the virtual structure of the tested compounds. In toxicology, the application of machine learning (ML) approaches has become increasingly widespread. These approaches enable the evaluation of large numbers of compounds in a time-efficient manner without involving animals as test subjects, in contrast to traditional in vivo methods [9].

Several researchers have conducted studies using machine learning (ML) to predict the toxicity of various compounds. In 2020, Elizeta Semenova and colleagues developed a Bayesian neural network model to predict the toxicity of compounds that have potential to cause liver injury. Their findings demonstrated that neural network methods are flexible and can reduce uncertainty when predicting compound toxicity [10]. In 2019, Abdul Karim and colleagues conducted research on toxicity prediction using multimodal deep learning. This method utilized multiple types of homogeneous neural

networks and diverse data representations. Their proposed method achieved significantly higher accuracy, ranging from 0.84 to 0.88, on the IGC50 benchmark dataset [11].

In 2021, Abdul Karim and his colleagues conducted a study that developed a deep learning ensemble framework to quantitatively predict compound toxicity levels. This research achieved significant improvements in quantitative toxicity predictions by integrating four quantitative toxicity datasets: LD50, IGC50, LC50, and LC50-DM. Among these datasets, the method used demonstrated enhanced performance, achieving improvements of 5.46%, 16.67%, and 6.34% in root-mean-square errors; 6.41%, 11.80%, and 12.16% in mean absolute errors; and 5.21%, 7.36%, and 2.54% in coefficients of determination for three out of the four datasets [12]. Literature reviews indicate that the development of such methods has delivered promising results, yet the models may not generalize well across diverse datasets or chemical classes, limiting the applicability in real-world scenarios. Therefore, the use of Simulated Annealing (SA) algorithms combined with Long Short-Term Memory (LSTM) networks in this study aims to improve toxicity predictions while optimizing computational efficiency and applicability of toxicity predictions in real-world scenarios.

This study focuses on predicting the toxicity of androgen receptor ligand-binding domain (AR-LBD) compounds using the LSTM model optimized with SA. LSTM is a specialized recurrent neural network (RNN) with a unique design structure. It features self-looping mechanisms that enable precise and consistent control over extended time periods [13]. In this case, SA is applied as an optimization technique to identify the optimal parameters for the LSTM model. Simulated Annealing excels at identifying the best solutions while reducing uncertainties in toxicity prediction. The integration of these methods aims to deliver improved prediction accuracy and greater computational efficiency.

II. MATERIALS AND METHODS

A. Dataset

This study utilizes a dataset from the Tox21 Data Challenge 2014 for the purpose of conducting a toxicity case study. The dataset functions as both training and testing data for the NR AR-LBD case analysis. A total of 4,034 compounds are used for training data, while 1,009 compounds are used for testing data. Molecular structures within this dataset are represented in SMILES format. In the first step of preprocessing, these structures are transformed into binary data, with 0 denoting "Normal" and 1 signifying "Toxicity."

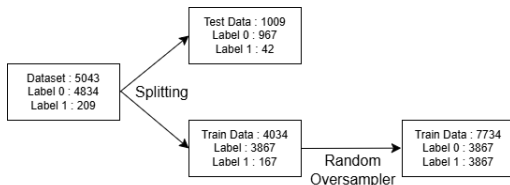


Fig. 1. Flow scheme of randomoversampler.

Fig. 1 presents the dataset distribution scheme from the initial stage to its preparation for use. The training set exhibits an imbalance between labels 0 and 1, indicating the necessity for oversampling via a Random Oversampler. As a result, there are 3,867 instances for both labels 0 and 1.

B. Long-Short Term Memory

The Long Short-Term Memory (LSTM) model is an advanced recurrent neural system specifically designed to overcome the vanishing gradient problem that often emerges during the learning of long-term dependencies. The LSTM architecture is composed of a series of recurrently connected sub-networks referred to as memory blocks. These memory blocks are designed to retain their state over time and manage the flow of information through non-linear gating units. The diagram in Fig. 2 illustrates the basic configuration of an LSTM block, which includes gates, input signal $x(t)$ output $y(t)$, activation functions, and peephole connections [14].

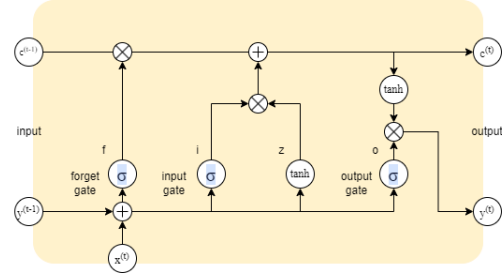


Fig. 2. Architecture block of LSTM.

The output of the block is recurrently connected back to the block's input and all gates. To explain the functionality of the LSTM model, consider a network consisting of N processing blocks and M inputs. Forward propagation within this recurrent neural framework is described as follows. Equation (1) is formulated based on the connections shown in Fig. 2, representing the mathematical expressions of the LSTM cell. [15].

$$\begin{aligned}
 f_t &= \omega(W_{fh}h_{t-1} + W_{fx}X_t + b_f), \\
 i_t &= \omega(W_{ih}h_{t-1} + W_{ix}X_t + b_i), \\
 \tilde{c}_t &= \tanh(W_{ch}h_{t-1} + W_{cx}X_t + b_{\tilde{c}}) \\
 c_t &= f_t \cdot c_{t-1} + i_t \cdot \tilde{c}_t, \\
 o_t &= \omega(W_{oh}h_{t-1} + W_{ox}X_t + b_o), \\
 h_t &= o_t \cdot \tanh(c_t).
 \end{aligned} \tag{1}$$

The forget gate in Fig. 2 determines what information should be discarded from the cell state. When the value of $f_t = 1$, the information is retained in the cell state. However, if $f_t = 0$, all information is discarded. The initialization of LSTM parameters is provided in Table I and Table II.

TABLE I
OPTIMIZED LSTM PARAMETER RANGE

Parameter	Value
Hidden Layers LSTM	[1,2,3]
Hidden Nodes LSTM	[1, ..., 128]
Activation Function	[tanh, relu, sigmoid]
Optimizer	[Adam, SGD, RMSprop]
Batch Size	[16, 32, 64]
Hidden Num Dense	[1, ..., 150]

C. Baseline

The goal is to establish a basic or baseline model for comparison in this study. Three baseline models were developed as the parameters were initialized in Table III. Baseline 1

TABLE II
FIXED LSTM PARAMETER

Parameter	Value
Validation Split	0.2
Callback	early stopping
Monitor	val_loss
Patience	10

includes a single LSTM layer with 128 nodes, relu activation, Adam optimizer, and a dense layer with 1 node. Baseline 2 comprises two LSTM layers, each with 128 nodes, relu activation, Adam optimizer, and a dense layer with 1 node. Baseline 3 features three LSTM layers, each with 128 nodes, relu activation, Adam optimizer, and a dense layer with 1 node.

TABLE III
LSTM BASELINE PARAMETER

Baseline	Hidden Layer	Units	Activation	Optimization	Batch Size
1	1	[128]	relu	Adam	32
2	2	[128, 128]	relu	Adam	32
3	3	[128, 128, 128]	relu	Adam	32

D. Simulated Annealing

Simulated Annealing (SA) is a fundamental and widely recognized meta-heuristic method for addressing complex global optimization challenges, particularly black-box problems where the objective function is not explicitly provided and requires computationally expensive simulations for evaluation [16]. This method relies on annealing techniques aimed at achieving the ground state of a material, which is the state with the lowest energy in a solid. In this context, when attempting to grow a single crystal from a melt (already liquefied), the lowest temperature proves inadequate for reaching the material's ground state.

The following Equation (2) represents the probability of an actual value x given by the Boltzmann distribution:

$$p(x) = e^{-\frac{\Delta f(x)}{kT}} \quad (2)$$

with $f(x)$ representing the energy configuration, k denoting the Boltzmann constant, and T being the temperature [17]. The standard SA optimization procedure involves the following steps:

- 1) To optimize the system, an initial solution x_0 is selected randomly, and the objective function is then evaluated.
- 2) Initialization of the temperature T is a critical parameter in implementing Simulated Annealing. A higher initial temperature requires more iterations for convergence, while a very low temperature may overlook points with potential global optima.
- 3) The acceptance of a new solution $x_0 + \Delta_x$ as the updated solution depends on T . The functions $f(x_0 + \Delta_x)$ and $f(x_0)$ represent the objective functions of $x_0 + \Delta_x$ and x_0 , respectively.
- 4) The solution $x_0 + \Delta_x$ is accepted and replaces x_0 if $f(x_0 + \Delta_x) \leq f(x_0)$. Once the optimal solution is updated, proceed to the next step. However, $x_0 + \Delta_x$

can also be accepted with a probability based on the Boltzmann distribution if $f(x_0 + \Delta_x) > f(x_0)$.

- 5) The temperature T decreases as the search progresses. At the beginning of the search, the probability of accepting worse solutions is relatively high, and it gradually decreases over time.
- 6) When the termination criteria are met, the computation ends. If the criteria are not yet satisfied, steps 2 through 6 are repeated continuously until they are fulfilled.

In the SA Algorithm, three parameters are utilized. The initial temperature, cooling rate, and maximum iterations, as summarized in Table IV. Subsequently, five exploration strategies are executed, each corresponding to a different cooling rate configuration, as outlined in Table V. These strategies are then compared based on the performance of an LSTM model optimized using the best discovered hyperparameters during the SA process.

TABLE IV
SIMULATED ANNEALING PARAMETER

Parameter	Value
Initial Temperature	100
Max Iterations	20
Cooling rate	[0.9, 0.85, 0.8, 0.75, 0.7]

TABLE V
EXPERIMENTAL SCHEME

Scheme	Cooling Rate
1	0.9
2	0.85
3	0.8
4	0.75
5	0.7

E. Confusion Matrix

In this study, the performance of the model is evaluated using the values from the confusion matrix [18]. Table VI includes attributes with descriptions for this stage. True Positive (TP) represents scenarios where positive data is predicted as positive and classified correctly. True Negative (TN) indicates that negative data is predicted as negative and classified correctly. Meanwhile, False Positive (FP) and False Negative (FN) refer to cases where positive and negative data are predicted as positive or negative, respectively, but classified incorrectly.

TABLE VI
CONFUSION MATRIX

Confusion Matrix	Prediction Condition		
		True (+)	False (-)
	Actual Condition	True (+)	False (-)
		TP	FN
		FP	TN

From this confusion matrix, several validation metrics can be calculated to measure the performance of the model,

including accuracy (3), F1-score (4), precision (5), and recall (6), using the following equations:

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \times 100\%, \quad (3)$$

$$F-1Score = \frac{2 \times (precision + recall)}{precision + recall} \times 100\%, \quad (4)$$

$$Precision = \frac{TP}{TP + FP} \times 100\%, \quad (5)$$

$$Recall = \frac{TP}{TP + FN} \quad (6)$$

III. RESULT AND DISCUSSION

A. Architecture Optimization

The analysis reveals that all five schemes show distinct convergence patterns across as shown in Fig. 3. Scheme 1 shows instability, with a significant drop at temperature 40 before stabilizing. Scheme 2 declines steadily until stabilizing, but a spike occurs at temperature 20. Scheme 3 exhibits the most stable behavior, maintaining loss values between 0.9 and 1.0 across temperatures. Scheme 4 performs well initially but becomes highly unstable below temperature 20. Scheme 5 shows a gradual decline and moderate stability, with lower convergence at mid-temperatures.

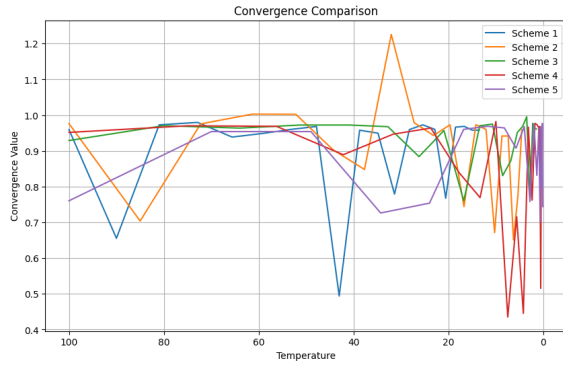


Fig. 3. Convergence plot.

Following the optimization phase of the architecture based on the predefined schemes outlined in Table V, the best parameters for each scheme were identified. These parameters include the number of hidden layers, number of nodes (Units), learning rates, activation, batch sizes, and optimizer.

The results of the LSTM parameter optimization through Simulated Annealing for each scheme are detailed in Table VII. For Scheme 1, with a cooling rate of 0.9, the optimal model consists of one hidden layer with 58 nodes, learning rate of 0.0049, using the tanh activation function, and the Adam optimizer. The batch size for this scheme is set to 16. Scheme 2, with a cooling rate of 0.85 and a learning rate of 0.0086, also features one hidden layer with node counts of 39, utilizing the relu activation function, the Adam optimizer, with a batch size of 16. The Scheme 3 has a cooling rate of 0.8 and a learning rate of 0.0093. The optimal model is characterized by a single layer, 59 nodes, the tanh activation function, the Adam optimizer, with a batch size of 16. With a cooling rate of 0.75 and a learning rate of 0.0053, the optimal model of Scheme 4 consists of one hidden layers with 40 nodes, employing the sigmoid activation function, the Adam optimizer, with a batch size of 16. Finally, for the cooling rate of 0.7 and a learning rate of 0.0056, the optimal model

of Scheme 5 features one hidden layer with 63 nodes, using the relu activation function, the Adam optimizer, and a batch size of 16.

TABLE VII
OPTIMIZED LSTM PARAMETER RESULT

Scheme	Hidden Layer	Units	Activation	Optimization	Batch Size
1	1	[58]	tanh	Adam	16
2	1	[39]	relu	Adam	16
3	1	[59]	tanh	Adam	16
4	1	[40]	sigmoid	Adam	16
5	1	[63]	relu	Adam	16

B. Validation Result

For this process, learning curve is presented in Fig. 4, serving as representative examples from the performance outcomes of the LSTM architecture. In these graphs, the loss and validation loss values obtained from both training and validation datasets are represented by X-axis, whereas the Y-axis indicates the epoch count. It is crucial to emphasize that the epoch counts vary for each architecture and do not reach a predetermined limit, which is a consequence of the early stopping callback implemented to cease training when significant improvements are no longer observed.

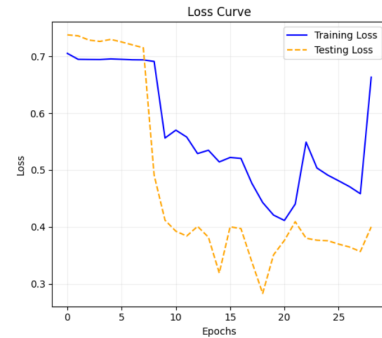


Fig. 4. Learning curve scheme 1.

Through the architecture optimization, a new LSTM model will be constructed based on the newly identified optimal parameters. The confusion matrix, as detailed in Table VI, will be used to carry out the evaluation process, with detailed results for both baseline and experimental scheme presented in Table VIII and Table IX respectively.

TABLE VIII
BASELINE RESULT FOR TRAIN AND TEST SET

Scheme	TP	TN	FP	FN	F-1 Score	Acc	Prec	Rec
Train								
Baseline 1	3333	2446	534	1421	0.7145	0.7472	0.8208	0.6325
Baseline 2	3800	506	67	3361	0.2279	0.5568	0.8831	0.1309
Baseline 3	3536	2315	331	1552	0.7109	0.7565	0.8749	0.5987
Test								
Baseline 1	874	33	93	9	0.3929	0.8989	0.2619	0.7857
Baseline 2	950	8	17	34	0.2388	0.9495	0.3200	0.1905
Baseline 3	910	32	57	10	0.4885	0.9336	0.3596	0.7619

Table VIII summarizes the confusion matrix results for both the train and test sets of baseline models that were predefined in Table III. On the train set, Baseline 1 obtained the best F1-score of 0.7145 and recall of 0.6325. Baseline 2 and Baseline 3 obtained the best accuracy of 0.7565 and precision of 0.8831 respectively on the train set. However, Baseline 2 outperformed other schemes with an accuracy of 0.9495 on the test set. In contrast, Baseline 1 recorded a TP count of 874 and a low F1-score of 0.3929, indicating its limitations in effectively classifying instances in the test set. Despite not achieving the best results in the other metrics, Baseline 3 achieved the highest TP count of 910 and lowest FN count of 50. Though each baseline excelled in different metrics, all three showed competitive results and consistent performance for both train and test sets.

TABLE IX
OPTIMIZATION RESULT FOR TRAIN AND TEST SET

Scheme	TP	TN	FP	FN	F-1 Score	Acc	Prec	Rec
Train								
Scheme 1	3652	2672	215	1195	0.7912	0.8177	0.9255	0.6910
Scheme 2	3783	2603	84	1264	0.7943	0.8257	0.9687	0.6731
Scheme 3	3831	3630	36	237	0.9638	0.9647	0.9902	0.9387
Scheme 4	3683	2863	184	1004	0.8282	0.8464	0.9396	0.7404
Scheme 5	3719	2324	148	1543	0.7332	0.7814	0.9401	0.6010
Test								
Scheme 1	917	32	50	10	0.5161	0.9405	0.3902	0.7619
Scheme 2	946	29	21	13	0.6304	0.9663	0.5800	0.6905
Scheme 3	960	27	7	15	0.7105	0.9782	0.7941	0.6429
Scheme 4	930	29	37	13	0.5370	0.9504	0.4394	0.6905
Scheme 5	941	27	26	15	0.5684	0.9594	0.5094	0.6429

As showed on the confusion matrix results for the training set in Table IX, Scheme 3 achieved the best result among the evaluated architecture with a TP count of 3,831, along with a FP count of 36, and a FN count of 237. This scheme achieved an impressive F1-score of 0.9638 and an accuracy of 0.9647, indicating its effectiveness in distinguishing between classes. Additionally, Scheme 2 performed admirably, recording a count TP count of 3,783, a FP count of 84, and F1-score of 0.7943. Table IX also provides the confusion matrix results for the test set, where Scheme 3 maintained its leading performance, achieving a TP of 960, an FN of 15, an F1-score of 0.7105, and an accuracy of 0.9782, indicating its strong generalization capability. Followed by Scheme 2 with a TP count of 946 and an F1-score of 0.6304. The comparison of each schemes' F1-score were outlined in Fig. 5 below.

Based on both model parameters and performance results obtained in Table IX and illustrated in Fig. 5, it can be concluded that Scheme 3 is the best architecture model for this SA-optimized LSTM model as it consistently emerged as the top-performing model across both training and testing datasets, achieving high accuracy and F1-score compared to other schemes with varying cooling rates. The results further support the use of SA for hyperparameter optimization, as Scheme 3 also outperforms baseline models where SA was not utilized. These findings indicate that the hyperparameter optimization process using SA is the key to enhance F1-score and accuracy. Scheme 3, which utilizes a cooling rate of 0.8 achieves the optimal combination of parameters, as the cool-

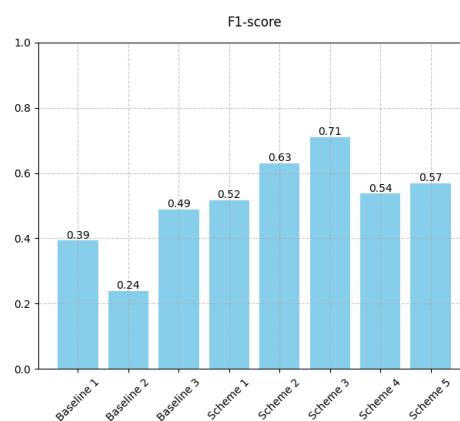


Fig. 5. F1-score comparison for each architecture.

ing rate establish an optimal balance between exploration and focused refinement during the hyperparameter optimization process as it allows the algorithm to explore a sufficient range of hyperparameter configurations with the best solutions and a more stable convergence pattern.

Additionally, Scheme 3 outperforms other schemes because its 39 units in a single hidden layer creates an optimal balance in model complexity without risking overfitting. The tanh activation function effectively prevents vanishing gradient during training, while Adam optimizer's adaptive learning rate helps find optimal weights efficiently. This balanced performance and optimal combination in Scheme 3 enable better and more stable performance for both training and test sets, indicating its strong generalization capabilities in handling the imbalanced classes in datasets. In contrast, other schemes lack this optimal combination, which contributes to their lower effectiveness in toxicity prediction as higher cooling rates may lead to excessive exploration while lower cooling rates may risk premature convergence.

IV. CONCLUSION

The implementation of an LSTM model optimized with Simulated Annealing (SA) for AR-LBD toxicity prediction has demonstrated promising results, particularly with Scheme 3, with an impressive an F1-score of 0.7105 and an accuracy of 0.9782 for the evaluation using the test data. Overall, this study highlights the potential of SA-optimized LSTM architectures for toxicity prediction, emphasizing the significance of optimization techniques in enhancing model performance. Despite the promising results of drug toxicity prediction from using the LSTM model optimized by SA, future work should focus on exploring additional hyperparameters and alternative optimization methods thoroughly to further enhance accuracy. Additionally, expanding the dataset to include a wider variety of compounds and toxicity could improve the model's applicability across diverse scenarios in toxicology.

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